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Experiment 5: Transient Couette Flow: Explicit and Implicit Formulations

1. Aim

To simulate non-dimensional 1D transient viscous momentum diffusion (plane Couette flow) using explicit and implicit Finite Volume Method (FVM) formulations based on Suhas V. Patankar's discretization approach, analyze their numerical convergence rates, and compare steady-state velocity profiles.

2. Problem Statement & Mathematical Model

Governing Equations & Non-Dimensionalization

The 1D unsteady Navier-Stokes momentum equation for incompressible viscous fluid flow between parallel plates separated by distance H without an applied pressure gradient is:

$$\frac{\partial u}{\partial t} = v\frac{\partial^2 u}{\partial y^2}$$

Introducing the non-dimensional spatial coordinate $y^* = y/H$, non-dimensional velocity$u^* = u/U$, non-dimensional time $t^* = t/T$, and the flow Reynolds number $Re = \frac{UH}{v}$ yields the governing equation:

$$\frac{\partial u^*}{\partial t^*} = (\frac{1}{Re})(\frac{TU}{H})\frac{\partial^2 u^*}{\partial y^{*2}}$$

Let $D = \left(\frac{1}{Re}\right)\left(\frac{TU}{H}\right)$represent the non-dimensional diffusion coefficient parameter.

3. Discretization Schemes (Finite Volume Method)

A. Explicit Scheme

Using central differencing for spatial diffusion terms evaluated at the previous time level n and forward differencing in time:

$$u_i^{n+1} = u_i^n + \Delta t.D[(u_{i+1}^n + u_{i-1}^n - 2u_i^n)/\Delta y^2]$$

Where the explicit time step is Δt_{step} defined via the CFL-like diffusion stability parameter F:

$$\Delta t_{step} = F.\Delta y^2.Re/(\frac{TU}{H})$$

For F = 0.5, Δy = 0.01, Re = 100, and T = U = H = 1:

$$\Delta t_{step} = 0.005\ sec$$

B. Implicit Scheme (Patankar's FVM Discretization)

Integrating the governing momentum equation over a 1D control volume of width Δy using in-place Gauss-Seidel updates at the current time level n+1:

$$u_i^{n+1} = \frac{(\frac{D}{\Delta y})(u_{i+1}^{n+1} + u_{i-1}^{n+1})(\frac{\Delta y}{\Delta t})u_i^n}{\left(\frac{2D}{\Delta y}\right) + (\frac{\Delta y}{\Delta t})}$$

Setting Δt_{imp}1.0 sec demonstrates the unconditional stability of the implicit FVM formulation

4. Boundary and Initial Conditions

- **Initial Condition:** u(y, 0) = 0 for all internal grid nodes ($0 \leq y \leq H$).
- **Bottom Boundary (y = 0):** u(0, t) = 0 (No-slip Dirichlet condition at stationary lower wall).
- **Top Boundary (y = H):** u(H, t) = U = 1.0 (No-slip Dirichlet condition at moving upper wall).

5. Input Parameters & Discretization Setup

Based on the numerical parameters defined in the source code:

Parameter Name	Variable	Value
Channel Height	H	1
Top Plate Speed	U	1
Reference Time	T	1
Reynolds Number	Re	100
Grid Resolution	Ny	100
Grid Spacing	Δy	0.01
Diffusion Parameter (F)	F	0.5
Explicit Time Step	Δt_{step}	0.005
Implicit Time Step	Δt_{imp}	1

```
"""
# A good resource to refer:
https://www.scribd.com/presentation/428957927/Couette-Flow-Solution-
Using-Analytic-and-Crank-nicolson-Numerical
```

```
Assumptions of Couette flow:
1. The flow is unsteady incompressible.
2. The flow is laminar and fully developed. (v=0)
3. No pressure gradient in the flow direction. (dp/dx=0)
4. Developed flow has no change in velocity in x direction. (du/dx=0)
5. Uniform properties (nu, rho)

Governing equation: du/dt = nu * d^2u/dy^2
Non dimensionalized: du*/dt* = (1/Re)*(TU/H) * d^2u*/dy*^2

Solved using FVM

No separate boundary treatment required for Dirichlet conditions
"""
import numpy as np
from matplotlib import pyplot as plt



ny = 100
H = 1
U = 1
T = 1
Re = 100
dy = H / ny
nit = 10000000
F = 0.5 # something like a CFL number
dt = F * dy**2 * Re / (T*U)

def apply_boundary(u):
    u[0] = 0
    u[-1] = U

# explicit solution
# initialization
u = np.zeros(ny)
u_old = u.copy()

def convergence_criteria(u, u_old):
    return np.max(np.abs(u - u_old))

for n in range(nit):
    apply_boundary(u)

    d2u_dy2 = (u_old[2:] + u_old[0:-2] - 2*u_old[1:-1])/(dy**2)
    non_dimensional = (1/Re)*(T*U/H)

    u[1:-1] = u_old[1:-1] + dt * (non_dimensional * d2u_dy2)
```

```
    if convergence_criteria(u, u_old) < 1e-6:
        print(f'Explicit converged after {n} iterations, dt = {dt}')
        exp_conv = n
        exp_dt = dt
        break

    u_old = u.copy()
u_explicit = u.copy()

# implicit solution
# initialization
u = np.zeros(ny)
u_old = u.copy()
dt = 1
for n in range(nit):
    apply_boundary(u)

    non_dimensional = (1/Re)*(T*U/H)
    u[1:-1] = ((non_dimensional/dy)*(u[2:]+u[0:-2]) + (dy/dt)*u_old[1:-
1])/((2*non_dimensional/dy) + (dy/dt)) # Patankar

    if convergence_criteria(u, u_old) < 1e-6:
        print(f'Implicit converged after {n} iterations, dt = {dt}')
        imp_conv = n
        imp_dt = dt
        break

    u_old = u.copy()
u_implicit = u.copy()

# plot results
fig, ax = plt.subplots(1, 2)
ax[0].plot(np.linspace(0, H, ny), u_explicit, label='Explicit
Solution')
ax[0].set_xlabel('Velocity (u)')
ax[0].set_ylabel('y')
ax[0].set_title(f'Explicit Solution - Converged after {exp_conv}
iterations, dt = {exp_dt}')
ax[1].plot(np.linspace(0, H, ny), u_implicit, label='Implicit
Solution')
ax[1].set_xlabel('Velocity (u)')
ax[1].set_ylabel('y')
ax[1].set_title(f'Implicit Solution - Converged after {imp_conv}
iterations, dt = {imp_dt}')
plt.show()
```

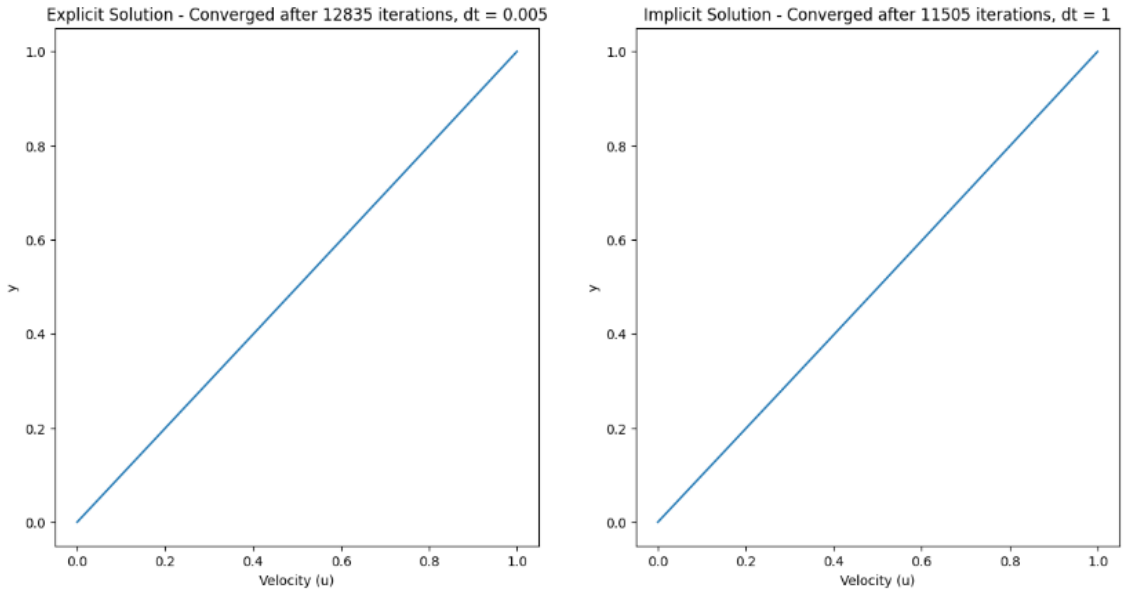
6. Results & Discussion

A. Numerical Convergence Summary

Both algorithms were executed until the maximum absolute velocity change between successive iterations satisfied

Numerical Formulation	Discretization Approach	Time Step	Iteration
Explicit Scheme	FTCS	0.005 sec	n=12835
Implicit Scheme	Patankar Point-Implicit	1 sec	n=11505

B. Velocity Profile & Stability Analysis



1. **Analytical Agreement:**
   - Both explicit and implicit FVM formulations converged cleanly to the exact theoretical steady-state velocity distribution for plane Couette flow:

$$u(y) = U \cdot \frac{y}{H}$$

   - The non-dimensional velocity varies linearly across the channel from u=0 at the lower wall (y=0) to u=1.0 at the upper moving boundary (y=1.0).
2. **Computational Performance & Time Step Sensitivity:**
   - **Explicit Scheme:** Bound by strict conditional stability constraints ($F \leq 0.5$), forcing the selection of a small time step ($\Delta t = 0.005$ s). It required **12,835 iteration steps** to reach thermal and momentum equilibrium.
   - **Implicit Scheme:** Leverages unconditional stability, allowing a time step 200 times larger ($\Delta t = 1.0$ s). Even when solved using a point-by-point Gauss-Seidel update rule without matrix inversion, it required **11,505 iteration steps**, demonstrating superior time-marching capability for steady-state seeking computations.

7. Conclusions

1. **Numerical Validation:** The Finite Volume Method formulations successfully resolve 1D non-dimensional transient Couette flow, producing matching linear velocity profiles at steady state.
2. **Explicit Stability Bounds:** The explicit FTCS formulation is constrained by diffusion stability limits ($F \leq 0.5$), requiring small physical time steps ($\Delta t = 0.005$ s) to prevent numerical divergence.
3. **Implicit Efficiency:** Patankar's implicit FVM formulation provides unconditional stability, allowing a significantly larger time step ($\Delta t = 1.0$ s) to achieve steady-state convergence efficiently.